

Hrvoje Jerković, PhD

Data Architect | Principal Data Engineer

Šibenik, Croatia | Remote B2B | European working hours

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Online CV | LinkedIn

Profile

Data and solution architect combining hands-on engineering, technical leadership and education. Data and ML project experience since 2017, with a broader background in information systems. Specialises in lakehouse modernisation, lineage, reliable pipelines and AI-assisted migration across financial services, consulting and the public sector.

Core expertise

- **Architecture and governance:** lakehouse/DWH, Medallion, relational and dimensional modelling, metadata, table/column lineage, data quality, access controls and auditability.
- **Engineering:** Python, SQL, dbt, Spark, Delta Lake, Airflow, Kafka, Snowflake, PostgreSQL and Oracle; batch, streaming, APIs and CDC patterns.
- **Cloud and delivery:** AWS, Google Cloud (GCP), Microsoft Azure, Databricks, DigitalOcean, Oracle Cloud, Docker, Git and CI/CD.
- **Analytics and marketing:** Google Analytics, e-marketing, cross-channel data integration, Tableau, Power BI and DAX.
- **Consulting and AI:** requirements discovery, architecture trade-offs, technical coordination, mentoring and workshops; AI-assisted ETL analysis and validated migration workflows.

Selected experience

Data Architect / Principal Data Engineer

Raiffeisenbank Austria d.d., Croatia (RBA) | Sep 2025 - Feb 2026 | Project case study

- Designed and implemented a lakehouse migration framework for an Oracle/PostgreSQL estate of approximately 4,300 ETL mappings and 1,800 landing tables. Defined Bronze/Silver/Gold layers and reusable dbt models targeting Spark and Delta Lake.
- Developed automated reverse engineering of ODI, DataStage and PL/SQL flows, reconstructing table- and column-level lineage across more than 20 source systems. Produced dependency graphs, SQL traceability and structured metadata for downstream automation.
- Built orchestration with nine specialised AI agents to translate legacy logic into dbt SQL. Ordered models by their dependencies, compiled and ran the SQL, compared results with production data and escalated failed checks.
- Removed the need for approximately 340 separately stored data-quality tables by running quality checks within the new processing flow. Handed over reusable models and pipelines to the delivery team.

Data Architect / Principal Data Engineer / Data Consultant

Intuita Consulting (UK group) | Sep 2022 - Feb 2024

- VIOOH / JCDecaux: led SQL/dbt model upgrades and maintenance in a Kafka-to-S3-to-Airflow-to-Snowflake pipeline handling approximately 1-5 million nested JSON records per hour. Investigated data issues using DataDog and Grafana.
- Designed and implemented a pipeline anomaly-detection and notification application using Python, SQL, AWS Athena, Lambda, Glue and Snowflake integrations.
- Altus Group: integrated Google Analytics, email campaigns, Salesforce, paid advertising and social-channel data into unified marketing analytics in Snowflake using dbt, Matillion and SQL. Built DataDQ advertising-data workflows with Airflow, S3 and Redshift.
- Comcast / Sky: developed a metadata extraction solution using Python, PostgreSQL and Collibra, and a separate tested GCP/AlloyDB prototype for metadata extraction and schema-change capture. Hyperoptic: mapped telecom data models to TM Forum SID/ODA using ER/UML modelling tools.

Further architecture, AI and education experience

Solution Architect / Technical Lead - eADR

Agency for Commercial Activities (AKD) / Croatian Ministry of the Sea, Transport and Infrastructure | Nov 2025 - Aug 2026

- Designed architecture and development specifications for six modules of an EU-funded national transport platform, connecting business processes, domain models, APIs, identity/RBAC, audit trails and reporting.
- Coordinated technical delivery and stakeholder decisions. Led user acceptance testing and integration checks, linked requirements to test cases, and defined API behaviour, error handling, retries and rules to prevent duplicate processing.

Data Engineering Instructor / Workshop Author

NEOS (KONČAR Group) | Apr 2026

- Designed and delivered a two-day Azure Databricks engineering workshop covering Delta Lake, Auto Loader, Structured Streaming, SQL, Unity Catalog, data quality and orchestration.
- Built banking demonstrations, guided labs and follow-up assignments, including dbt tests, Bundles and delivery workflows. Connected practical exercises with Associate/Professional certification topics and adapted material to the team's environment.

Lead Data Architect / Project Manager - Technical Bid

NEOS / Central Bank of Oman — pre-contract evaluation | Sep 2025 - Feb 2026

- Led solution design and technical coordination; developed a FastAPI/Vue demonstration application and prototypes for regulatory returns, submissions, revisions and analytics, alongside the proposed on-premises lakehouse architecture.
- Reached final-stage evaluation with the NEOS team and presented my work and project references to the bank's delegation in Zagreb. The engagement did not progress to contract award or production implementation.

Data Architect / Technical Lead - Road Transport Analytics

Agency for Commercial Activities (AKD) / Croatian Ministry of the Sea, Transport and Infrastructure | Jul - Dec 2024

- Designed an open-source lakehouse for a planned 250 TB+ scope covering 75,000+ drivers and 65,000+ vehicles. Defined 40+ safety, compliance and efficiency indicators integrating tachograph, inspection and ministry data.
- Prepared infrastructure and processing specifications, governance requirements and technical RFI/RFP inputs. Proposed Spark, MinIO, dbt, Dagster, OpenMetadata and PostgreSQL; aligned platform, engineering and analytics needs.

Data Architect / Head of AI

8tracks.com / mixer.fm | Apr - Aug 2024

- Developed AWS audio-processing components with S3, Lambda, EFS, PostgreSQL/RDS, SAM, CloudFormation and CodeBuild. Integrated pretrained models for genre, style, mood, instrumentation and danceability; built audio-feature extraction, metadata processing and an interactive waveform demonstration. Designed a recommendation-service architecture.

Data Engineer / Data Scientist / Lecturer

Algebra / Algebra LAB | Mar 2018 - Sep 2022

- IRI2 / Neural Grid: developed real-time software to compare phasor angles from PMU measurements, perform power-system calculations and trigger PLC actions for grid regulation. Used Python, Java and Kafka Streams; achieved a response below 50 ms, significantly faster than SCADA in this project.
- National labour-market analytics: migrated analytical processing to SQL Server and integrated employment and education data. Agricultural analytics: developed complex SQL views, a large Tableau Hyper extract and ministry-wide dashboards.
- Delivered data engineering and computer-vision projects, including Erasmus ADSEE. Course leader for Machine Learning on the Data Science master's, delivering part of its lectures and practical exercises; also taught Disruptive Technologies and E-business.

Earlier experience: Zagreb School of Economics and Management, 2002-2017 - information systems development and integration, technical leadership, digital learning and professional education.

Additional project work

- Vehicle-market analytics and price prediction: developed an XGBoost car-price model for Oglasnik.hr, with Tableau and Power BI analytics. Built an independent Airflow/PostgreSQL collection platform with new-ad detection and price-change tracking.
- Desktop property analytics: developed a Python/PyQt5 application for collecting, filtering and analysing property listings, with configurable notifications.
- News portal: built Python/PostgreSQL collection, Dagster jobs, AI summaries and article/category publishing; developed the project using Oracle Cloud.
- National register of regulated professions and occupational classifications: built SQL Server databases and C#/ASP.NET Core applications for nationwide reference-data maintenance, search, editing and audit.
- eFTI portal: developed Java/Spring Boot and React/TypeScript software for freight datasets, XML validation, subset selection, access links and audit.

Education and professional development

PhD in Applied Computer Science — University of Zagreb, Faculty of Electrical Engineering and Computing

Awarded 2012. Research: information-system interoperability and integration of learning platforms with digital repositories.

Udacity: Machine Learning Engineer Nanodegree; Deep Learning Nanodegree.

Credentials: SnowPro Core (2024); AWS Cloud Practitioner (2022).

Languages: English - full professional proficiency, including consulting and teaching; Croatian - native.

Public Databricks workshop materials